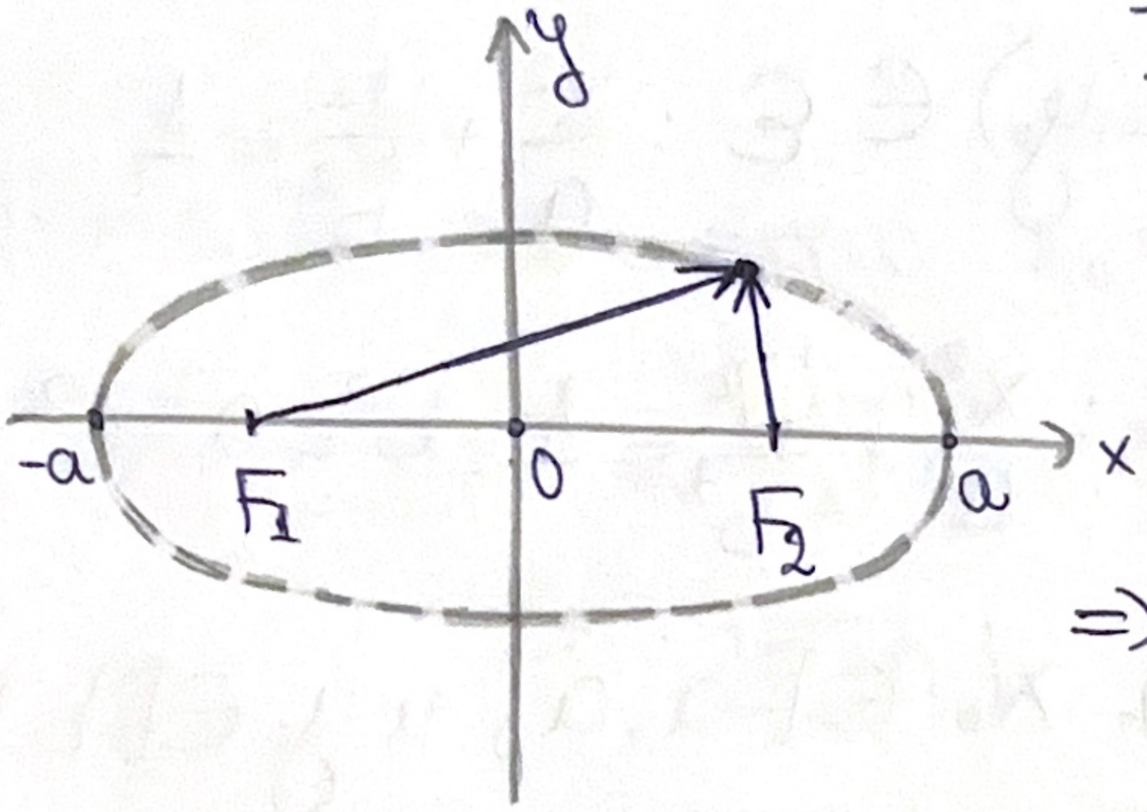
Упражнение № 11

Конични сечения - основни свойства

1 зад. (Основна задача - Фокални св-ства).

а) Нека F_1 и F_2 са две различни точки. $|F_1F_2| = 2c$. Дадено е $a > c$.

Търсим ГМ на $\forall$ т. M от равнината, за които $|\overrightarrow{F_1M}| + |\overrightarrow{F_2M}| = 2a$.



I Избор на ОКС $K = Oxy$

1) т. O - среда на F_1F_2

2) $Ox \equiv F_1F_2$, $Ox^+ \uparrow\uparrow \overrightarrow{F_1F_2}$

3) Oc $Oy \equiv \beta_{F_1F_2}$

$\Rightarrow F_1(-c, 0), F_2(c, 0)$

$|\overrightarrow{F_1F_2}| = 2c$, $\overrightarrow{F_1F_2}(2c, 0)$,

$0 < c < a$ спрямо K

$\Rightarrow$ II Нека т. $M_0(x_0, y_0)$ спрямо K и нека $|\overrightarrow{F_1M_0}| + |\overrightarrow{F_2M_0}| = 2a$ (1), $\overrightarrow{F_1M_0}(x_0 + c, y_0)$, $\overrightarrow{F_2M_0}(x_0 - c, y_0)$

ще намерим у-е на фигурата, която описва всички точки M_0 с това свойство.

Заместване в (1):

$$|\overrightarrow{F_1M_0}| + |\overrightarrow{F_2M_0}| = 2a$$

$$\| \qquad \|$$

$$\sqrt{(x_0+c)^2 + y_0^2} + \sqrt{(x_0-c)^2 + y_0^2} = 2a$$

$$\sqrt{(x_0+c)^2 + y_0^2} = 2a - \sqrt{(x_0-c)^2 + y_0^2} \quad \uparrow^2$$

$$x_0^2 + 2x_0c + c^2 + y_0^2 = 4a^2 - 4a\sqrt{(x_0-c)^2 + y_0^2} + x_0^2 - 2x_0c + c^2 + y_0^2$$

$$4x_0c = 4a^2 - 4a\sqrt{(x_0-c)^2 + y_0^2} \quad /:4$$

$$x_0c = a^2 - a\sqrt{(x_0-c)^2 + y_0^2}$$

$a\sqrt{(x_0-c)^2+y_0^2} = a^2 - x_0c \uparrow^2$

$a^2(x_0^2 - 2x_0c + c^2 + y_0^2) = a^4 - 2a^2x_0c + x_0^2c^2$

$\underline{a^2x_0^2} - \cancel{2a^2x_0c} + \underline{a^2c^2} + a^2y_0^2 = \underline{a^4} - \cancel{2a^2x_0c} + \underline{x_0^2c^2}$

$x_0^2(a^2-c^2) + a^2y_0^2 = a^2(a^2-c^2)$

Понаречие $\underline{b^2 = a^2-c^2}$: $(a^2-b^2=c^2)$

$b^2 \cdot x_0^2 + a^2 \cdot y_0^2 = a^2 \cdot b^2 \quad /:(a^2 \cdot b^2)$

$\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} = \underline{1} \Rightarrow$ т. $M_0(x_0,y_0) \in \mathcal{E} : \frac{x^2}{a^2} + \frac{y^2}{b^2} = \underline{1}$

($\Leftarrow$) III Нека т. $M_0(x_0,y_0) \in \mathcal{E} : \frac{x^2}{a^2} + \frac{y^2}{b^2} = \underline{1}$, $b^2 = a^2-c^2$

$\Rightarrow \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} = \underline{1}$ $(b^2=a^2-c^2)$, $x_0 \in [-a,a]$ и $y_0 \in [-b,b]$

Ще докажем, че $|\overrightarrow{F_1M_0}| + |\overrightarrow{F_2M_0}| = 2a$.

$|\overrightarrow{F_1M_0}| = \sqrt{(x_0+c)^2+y_0^2} \uparrow^2$

$|\overrightarrow{F_2M_0}| = \sqrt{(x_0-c)^2+y_0^2} \uparrow^2$

$|\overrightarrow{F_1M_0}|^2 = x_0^2 + 2x_0c + c^2 + y_0^2$

$|\overrightarrow{F_2M_0}|^2 = x_0^2 - 2x_0c + c^2 + y_0^2$

От $\frac{x_0^2}{a^2} + \frac{y_0^2}{a^2-c^2} = \underline{1} \Rightarrow y_0^2 = \left(\underline{1} - \frac{x_0^2}{a^2}\right) \cdot (a^2-c^2) = \frac{\underline{(a^2-x_0^2)(a^2-c^2)}}{a^2}$

$\Rightarrow |\overrightarrow{F_1M_0}|^2 = x_0^2 + 2x_0c + \cancel{c^2} + a^2 - \cancel{c^2} - \frac{\underline{x_0^2(a^2-c^2)}}{a^2}$

$|\overrightarrow{F_2M_0}|^2 = x_0^2 - 2x_0c + c^2 + a^2 - c^2 - \frac{\underline{x_0^2(a^2-c^2)}}{a^2}$

$|F_1M|^2 = \frac{c^2}{a^2}x_0^2 + 2x_0c + a^2 = \left(\frac{c}{a}x_0 + a\right)^2$

$|\overrightarrow{F_2M}|^2 = \frac{c^2}{a^2}x_0^2 - 2x_0c + a^2 = \left(\frac{c}{a}x_0 - a\right)^2$

$\Rightarrow |\overrightarrow{F_1M_0}| = \left|\frac{c}{a}x_0 + a\right| = \left|\frac{cx_0 + a^2}{a}\right|$

$|\overrightarrow{F_2M_0}| = \left|\frac{c}{a}x_0 - a\right| = \left|\frac{cx_0 - a^2}{a}\right|$

$c < a$

$|x_0| \le a$, защото $x_0 \in [-a, a]$

$\Rightarrow |cx_0| < a^2 \Rightarrow |\overrightarrow{F_1M_0}| = \frac{a^2 + cx_0}{a}$

$|\overrightarrow{F_2M_0}| = \frac{a^2 - cx_0}{a}$

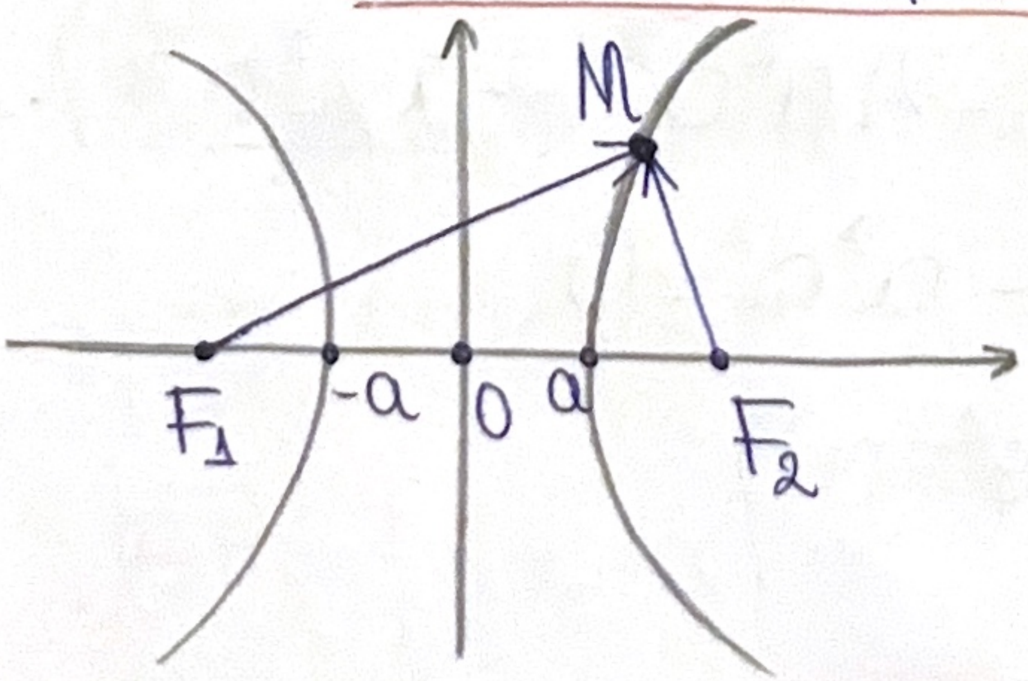
$\Rightarrow |\overrightarrow{F_1M_0}| + |\overrightarrow{F_2M_0}| = \frac{a^2 + \cancel{cx_0}}{a} + \frac{a^2 - \cancel{cx_0}}{a} = \frac{2a^2}{a} = 2a$

от II и III $\Rightarrow$ ГМ на $\forall$ т. M от равнината, за която $|F_1M| + |F_2M| = 2a$ е елипса

фокално св-во на елипса

б) $F_1 \neq F_2$, $|F_1F_2| = 2c$, $a \in (0; c)$, т.е. $0 < a < c$

Търсим ГМ на $\forall$ т. M от равнината, за която $||F_1M| - |F_2M|| = 2a$.



I Избор на ОКС $K = Oxy$

1) т. O - среда на F_1F_2

2) $Ox \equiv F_1F_2$, $Ox^+ \uparrow\uparrow F_1F_2$

3) $Oy \equiv S_{F_1F_2}$

$\Rightarrow F_1(-c, 0), F_2(c, 0)$

(⇒) Нека т. $M_0(x_0,y_0)$ спрямо K и нека

$||\overrightarrow{F_1M_0}|-|\overrightarrow{F_2M_0}||=2a$

$\overrightarrow{F_1M_0}(x_0+c,y_0)$, $\overrightarrow{F_2M_0}(x_0-c,y_0)$ спрямо K

$|\overrightarrow{F_1M_0}|=\sqrt{(x_0+c)^2+y_0^2}$, $|\overrightarrow{F_2M_0}|=\sqrt{(x_0-c)^2+y_0^2}$

$\Rightarrow\left|\sqrt{(x_0+c)^2+y_0^2}-\sqrt{(x_0-c)^2+y_0^2}\right|=2a\ \uparrow^2$

$x_0^2+2x_0c+c^2+y_0^2-2\sqrt{(x_0+c)^2+y_0^2}\cdot\sqrt{(x_0-c)^2+y_0^2}$

$+x_0^2-2x_0c+c^2+y_0^2=4a^2\ /:2$

$x_0^2+y_0^2+c^2-\sqrt{(x_0+c)^2+y_0^2}\sqrt{(x_0-c)^2+y_0^2}=2a^2$

$x_0^2+y_0^2+c^2-2a^2=\sqrt{(x_0+c)^2+y_0^2}\cdot\sqrt{(x_0-c)^2+y_0^2}\ \uparrow^2$

$x_0^4+y_0^4+c^4+4a^4+2x_0^2y_0^2+2x_0^2c^2-4a^2x_0^2+2y_0^2c^2$

$-4a^2y_0^2-4a^2c^2=(x_0^2+2x_0c+c^2+y_0^2)(x_0^2-2x_0c$

$+c^2+y_0^2)$

$x_0^4+y_0^4+c^4+4a^4+2x_0^2y_0^2+2x_0^2c^2-4a^2x_0^2+2y_0^2c^2-4a^2y_0^2$

$-4a^2c^2=$

$=x_0^4-2x_0^3c+x_0^2c^2+x_0^2y_0^2+2x_0c-4x_0^2c^2+2x_0c^3+2x_0cy_0^2$

$+c^2x_0^2-2x_0c^3+c^4+c^2y_0^2+y_0^2x_0^2-2x_0cy_0^2+c^2y_0^2+y_0^4$

$\Rightarrow 4a^4+2x_0^2c^2-4a^2x_0^2-4a^2y_0^2-4a^2c^2=-2x_0^2c^2\ /:4$

$a^4+x_0^2c^2-a^2x_0^2-a^2y_0^2-a^2c^2=0$

$x_0^2(c^2-a^2)-a^2y_0^2=a^2(c^2-a^2)$

Полагаме $b^2=c^2-a^2$:

$x_0^2 \cdot b^2 - a^2 y_0^2 = a^2 \cdot b^2 \;/:(a^2 b^2)$

$\frac{x_0^2}{b^2} - \frac{y_0^2}{a^2} = 1 \Rightarrow$ т. $M_0(x_0, y_0) \in \chi : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

($\Leftarrow$) III Нека т. $M_0(x_0, y_0) \in \chi : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $b^2 = c^2 - a^2$

$\Rightarrow \frac{x_0^2}{a^2} - \frac{y_0^2}{b^2} = 1$, $x_0 \in (-\infty; -a] \cup [a; +\infty)$, $y \in \mathbb{R}$

Ще док., че $||\overrightarrow{F_1M}| - |\overrightarrow{F_2M}|| = 2a$

$$\left.\begin{array}{l} |\overrightarrow{F_1M}_0| = \sqrt{(x_0+c)^2 + y_0^2}\,\uparrow^2 \\ |\overrightarrow{F_2M}_0| = \sqrt{(x_0-c)^2 + y_0^2}\,\uparrow^2 \end{array}\right| \Rightarrow \left|\begin{array}{l} |\overrightarrow{F_1M}_0|^2 = x_0^2 + 2x_0c + c^2 + y_0^2 \\ |\overrightarrow{F_2M}_0|^2 = x_0^2 - 2x_0c + c^2 + y_0^2 \end{array}\right.$$

Но $\frac{x_0^2}{a^2} - \frac{y_0^2}{b^2} = 1 \Rightarrow y_0^2 = \left(\frac{x_0^2}{a^2} - 1\right) b^2$, $b^2 = c^2 - a^2$

$\Rightarrow y_0^2 = \frac{(x_0^2 - a^2)}{a^2} \cdot (c^2 - a^2)$

$$\Rightarrow \left|\begin{array}{l} |\overrightarrow{F_1M}_0| = x_0^2 + 2x_0c + c^2 + \frac{(x_0^2 - a^2)}{a^2}(c^2 - a^2) \\ |\overrightarrow{F_2M}_0| = x_0^2 - 2x_0c + c^2 + \frac{(x_0^2 - a^2)}{a^2}(c^2 - a^2) \end{array}\right.$$

$$\left|\begin{array}{l} |\overrightarrow{F_1M}_0| = \frac{c^2}{a^2}x_0^2 + 2x_0c + a^2 = \left(\frac{c}{a}x_0 + a\right)^2 \\ |\overrightarrow{F_2M}_0| = \frac{c^2}{a^2}x_0^2 - 2x_0c + a^2 = \left(\frac{c}{a}x_0 - a\right)^2 \end{array}\right.$$

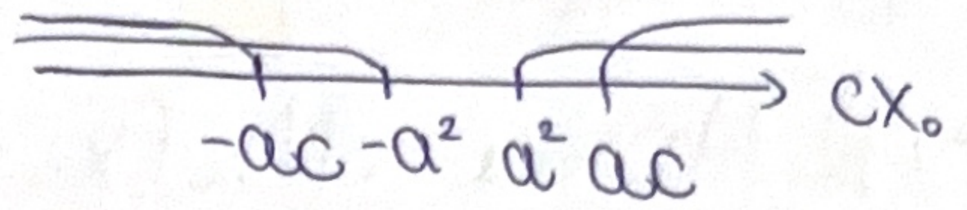
$$\Rightarrow \left|\begin{array}{l} |\overrightarrow{F_1M}_0| = \left|\frac{c}{a}x_0 + a\right| \\ |\overrightarrow{F_2M}_0| = \left|\frac{c}{a}x_0 - a\right| \end{array}\right.$$

Но $\left|\begin{array}{l} 0 < a < c \\ x_0 \in (-\infty; -a] \cup [a; +\infty) \end{array}\right.$

$|\overrightarrow{F_1M_0}| = \left|\frac{cx_0+a^2}{a}\right|$, $|\overrightarrow{F_2M_0}| = \left|\frac{cx_0-a^2}{a}\right|$

$x_0 \in (-\infty;-a] \cup [a;+\infty)$

$cx_0 \in (-\infty;-ac) \cup (ac;+\infty)$



$0<a<c \Rightarrow |cx_0| > a^2$

$\Rightarrow |\overrightarrow{F_1M_0}| = \frac{cx_0+a^2}{a}$, $|\overrightarrow{F_2M_0}| = \frac{cx_0-a^2}{a}$

$\left||\overrightarrow{F_1M_0}| - |\overrightarrow{F_2M_0}|\right| = \frac{cx_0+a^2}{a} - \frac{cx_0-a^2}{a} = 2a$

от II и III $\Rightarrow$ ГМ на $\forall$ т. M_0 от равнината, за която $||\overrightarrow{F_1M_0}| - |\overrightarrow{F_2M_0}|| = 2a$ е хипербола.

В) дадени са т. F и права g, $F \notin g$; $d(F;g) =$

Търсим ГМ на $\forall$ т. M, за която

$|\overrightarrow{FM}| = d(M,g)$

I Избор на ОКС $K = Oxy$

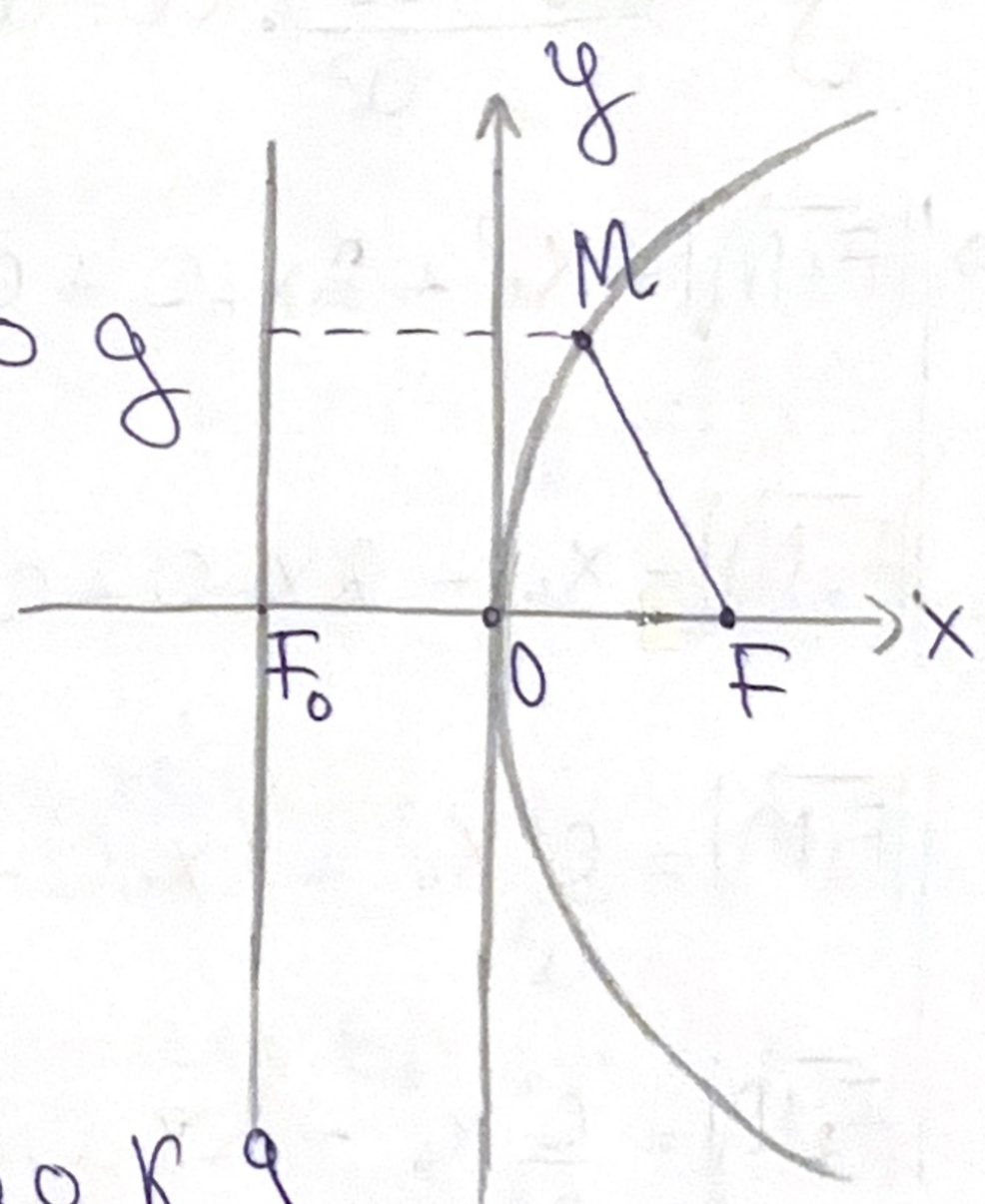
1) F_0 е орт. пр. на F относно g

2) O е среда на FF_0

3) $Ox \equiv F_0F$, $Ox^+ \uparrow\uparrow F_0F$

4) $Oy \equiv S_{F_0F}$

$\Rightarrow F\left(\frac{p}{2};0\right)$ и $g: x = -\frac{p}{2}$



II ($\Rightarrow$) Нека т. $M_0(x_0,y_0)$ спрямо K g и нека $|\overrightarrow{FM_0}| = d(M_0,g)$

$\overrightarrow{FM_0}\left(x_0-\frac{p}{2}, y_0\right) \Rightarrow |\overrightarrow{FM_0}| = \sqrt{\left(x_0-\frac{p}{2}\right)^2 + y_0^2}$

$d(M_0,g) = x_0 + \frac{p}{2}$

$\Rightarrow \sqrt{(x_0-\frac{p}{2})^2+y_0^2} = x_0+\frac{p}{2} \uparrow^2$

$(x_0-\frac{p}{2})^2+y_0^2 = (x_0+\frac{p}{2})^2$

$\frac{p^2}{4} + x_0^2 - x_0p + y_0^2 = x_0^2 + x_0p + \frac{p^2}{4}$

$\Rightarrow y_0^2 = 2x_0p \Rightarrow M_0 \in \Pi : y^2 = 2px$

III($\Leftarrow$) Нека т. $M_0(x_0,y_0) \in \Pi : y^2 = 2px$

$\Rightarrow y_0^2 = 2p.x_0$ (*)

$|\overrightarrow{FM_0}| = \sqrt{(x_0-\frac{p}{2})^2+y_0^2}$

$|\overrightarrow{FM_0}|^2 = x_0^2 - px_0 + \frac{p^2}{4} + y_0^2 \Rightarrow |\overrightarrow{FM_0}|^2 = x_0^2 - px_0 + \frac{p^2}{4} + 2px_0$

от (*)

$\Rightarrow |\overrightarrow{FM_0}|^2 = x_0^2 + px_0 + \frac{p^2}{4} \Rightarrow |\overrightarrow{FM_0}| = (x_0+\frac{p}{2})^2$

$\Rightarrow |\overrightarrow{FM_0}| = |x_0+\frac{p}{2}|$, $x_0 > 0 \Rightarrow |\overrightarrow{FM_0}| = x_0+\frac{p}{2} = d(M_0,g)$

от II и III $\Rightarrow$ ГМ на $\forall$ т. M_0 от равнината, за която $|\overrightarrow{FM_0}| = d(M_0,g)$ е парабола.

2зад. ОКС $K = Oxy$

Права $g: y = kx+n$ е допирателна към

а) $\mathcal{E}: \frac{x^2}{a^2}+\frac{y^2}{b^2} = 1 \Leftrightarrow a^2.k^2 = n^2-b^2$

б) $\chi: \frac{x^2}{a^2}-\frac{y^2}{b^2} = 1 \Leftrightarrow a^2.k^2 = n^2+b^2$

в) $\Pi: y^2 = 2px \Leftrightarrow \begin{matrix} p = 2k.n \\ k \neq 0 \end{matrix}$

доК.: а) $\varepsilon \cap g = \{T\} \Rightarrow$

$\left|\begin{array}{l} \frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \\ y=kx+n \end{array}\right. \Rightarrow \frac{x^2}{a^2}+\frac{(kx+n)^2}{b^2}=1$

$b^2x^2+a^2(k^2x^2+2knx+n^2)=a^2b^2$

$x^2(b^2+a^2k^2)+x\cdot 2kna^2+a^2n^2-a^2b^2=0$

$D=(kna^2)^2-(b^2+a^2k^2)(a^2n^2-a^2b^2)$

$D=\cancel{k^2n^2a^4}-a^2b^2n^2+a^2b^4-\cancel{a^4k^2n^2}$
$+a^4b^2k^2=$

$=a^2b^2(-n^2+b^2+a^2k^2)=0$

$\Rightarrow -n^2+b^2+a^2k^2=0$

$\Rightarrow n^2=b^2+a^2k^2$

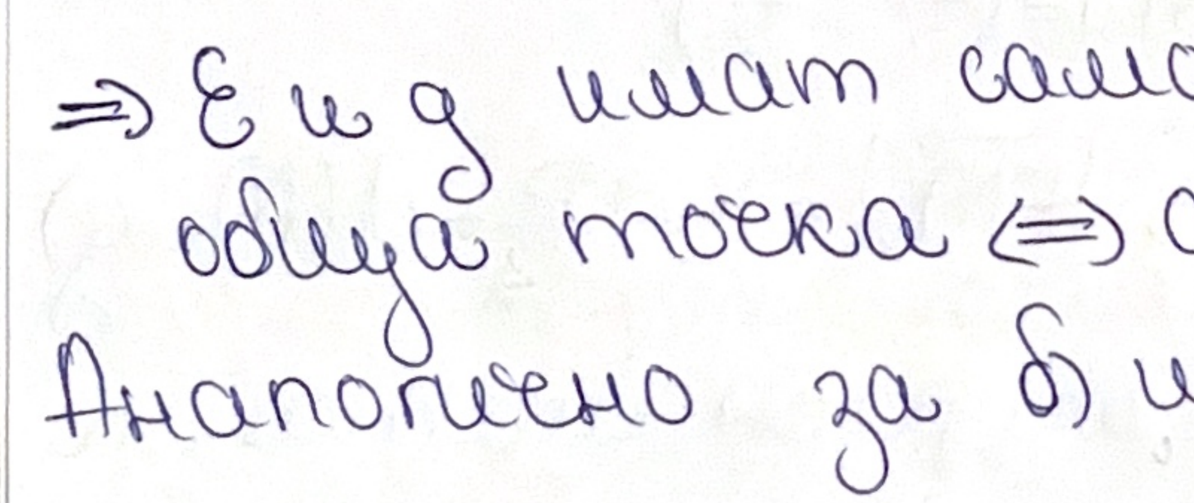
За да имат само 1 обща точка!

$\Rightarrow$ ε и g имат само една обща точка $\Leftrightarrow a^2k^2=n^2-b^2$

Аналогично за б) и в). ■

Задач. (Оптични свойства)

а) Ако ε:



ОКС $R=Oxy$

$\varepsilon: \frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

$c^2=a^2-b^2$

$M_0(x_0,y_0)\in\varepsilon$

$t_0: \frac{x\cdot x_0}{a^2}+\frac{y\cdot y_0}{b^2}=1$ - допирателна в т. M_0 към ε

$t_0 \cap Ox = N$

Ще докажем, че t_0 е външната ъглополовяща при M_0 на $\Delta F_1F_2M_0$, т.е. ще докажем, че

$$\frac{|\overrightarrow{F_1N}|}{|\overrightarrow{F_2N}|} = \frac{|\overrightarrow{F_1M_0}|}{|\overrightarrow{F_2M_0}|}$$

док.: $F_1(-c,0)$, $F_2(c,0)$

$t_0 \cap Ox = т.N \Rightarrow N\left(\frac{a^2}{x_0}, 0\right)$

$\overrightarrow{F_1N}\left(\frac{a^2}{x_0}+c, 0\right)$, $\overrightarrow{F_2N}\left(\frac{a^2}{x_0}-c, 0\right) \Rightarrow$

$|\overrightarrow{F_1N}| = \left|\frac{a^2}{x_0}+c\right|$, $|\overrightarrow{F_2N}| = \left|\frac{a^2}{x_0}-c\right| \Rightarrow \frac{|\overrightarrow{F_1N}|}{|\overrightarrow{F_2N}|} = \frac{\left|\frac{a^2+cx_0}{x_0}\right|}{\left|\frac{a^2-cx_0}{x_0}\right|} \frac{|a^2+cx_0|}{|a^2-cx_0|}$ (1)

$\overrightarrow{F_1M_0}(x_0+c, y_0)$, $\overrightarrow{F_2M_0}(x_0-c, y_0)$

$|\overrightarrow{F_1M_0}| = \sqrt{(x_0+c)^2+y_0^2}$, $|\overrightarrow{F_2M_0}| = \sqrt{(x_0-c)^2+y_0^2}$

$M_0 \in \mathcal{E}: \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} = 1$ и $a^2-b^2=c^2$

$\Rightarrow y_0^2 = \left(1-\frac{x_0^2}{a^2}\right)b^2$

$|\overrightarrow{F_1M_0}|^2 = x_0^2 + 2x_0c + c^2 + y_0^2 =$

$= x_0^2 + 2x_0c + c^2 + \left(a_2-x_0^2\right)\cdot(a^2-c^2) =$

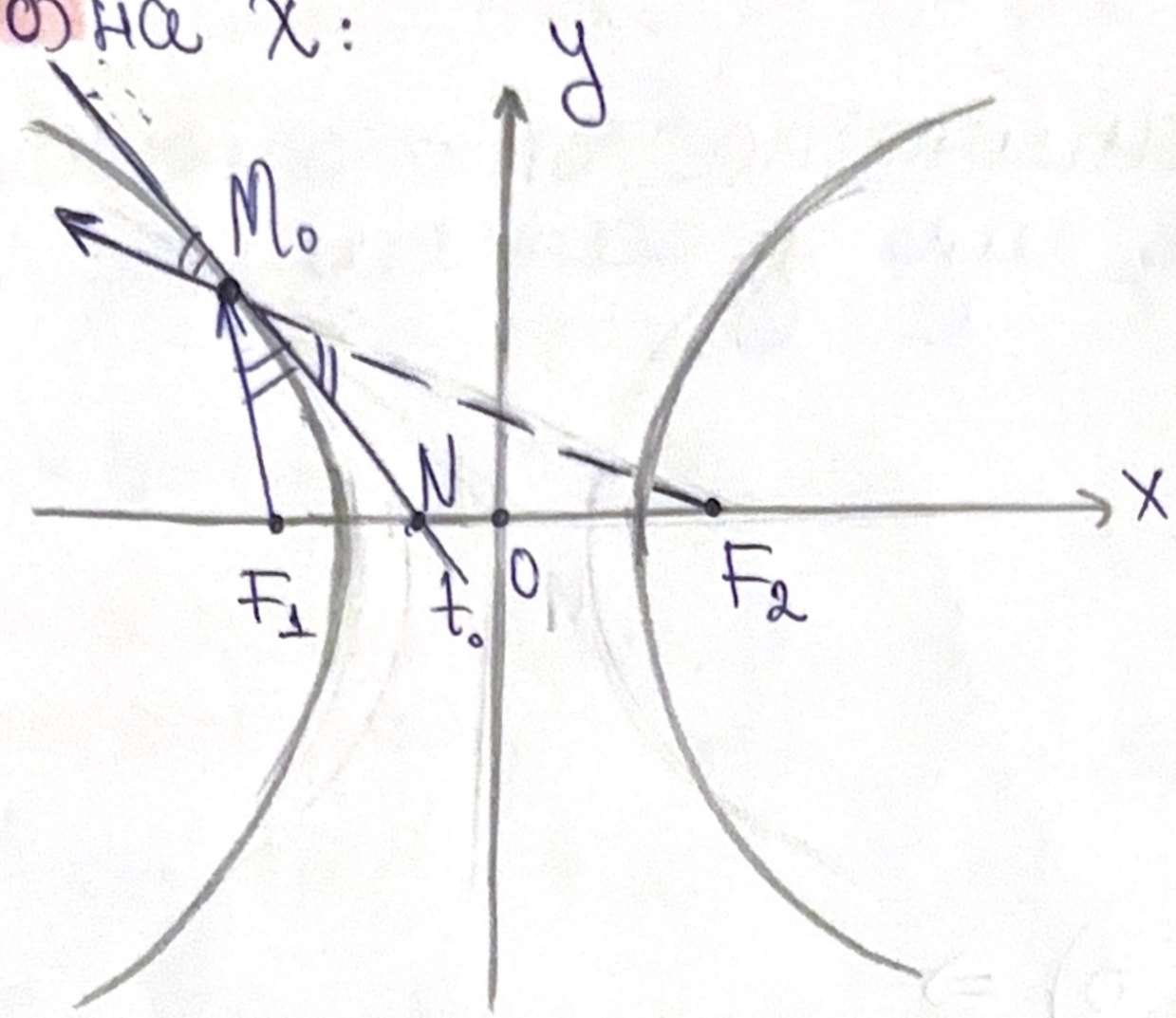
$= \frac{c^2}{a^2}x_0^2 + 2x_0c + a^2 = \left(\frac{c}{a}x_0+a\right)^2$

$\Rightarrow |\overrightarrow{F_1M_0}| = \left|\frac{a^2+cx_0}{a}\right|$

Аналог. $|\overrightarrow{F_2M_0}| = \left|\frac{a^2-cx_0}{a}\right|$ $\Rightarrow \frac{|\overrightarrow{F_1M_0}|}{|\overrightarrow{F_2M_0}|} = \frac{|a^2+cx_0|}{|a^2-cx_0|}$ (2)

От (1) = (2) $\Rightarrow \frac{|\overrightarrow{F_1N}|}{|\overrightarrow{F_2N}|} = \frac{|\overrightarrow{F_1M_0}|}{|\overrightarrow{F_2M_0}|} \Rightarrow$ t_0 е външната ъглополовяща

б) на χ:



ОКС $K=Oxy$

$\chi: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $c^2 = a^2 + b^2$

$F_1(-c,0)$, $F_2(c,0)$

$M_0(x_0,y_0) \in \chi$

$t_0: \frac{x \cdot x_0}{a^2} - \frac{y \cdot y_0}{b^2} = 1$

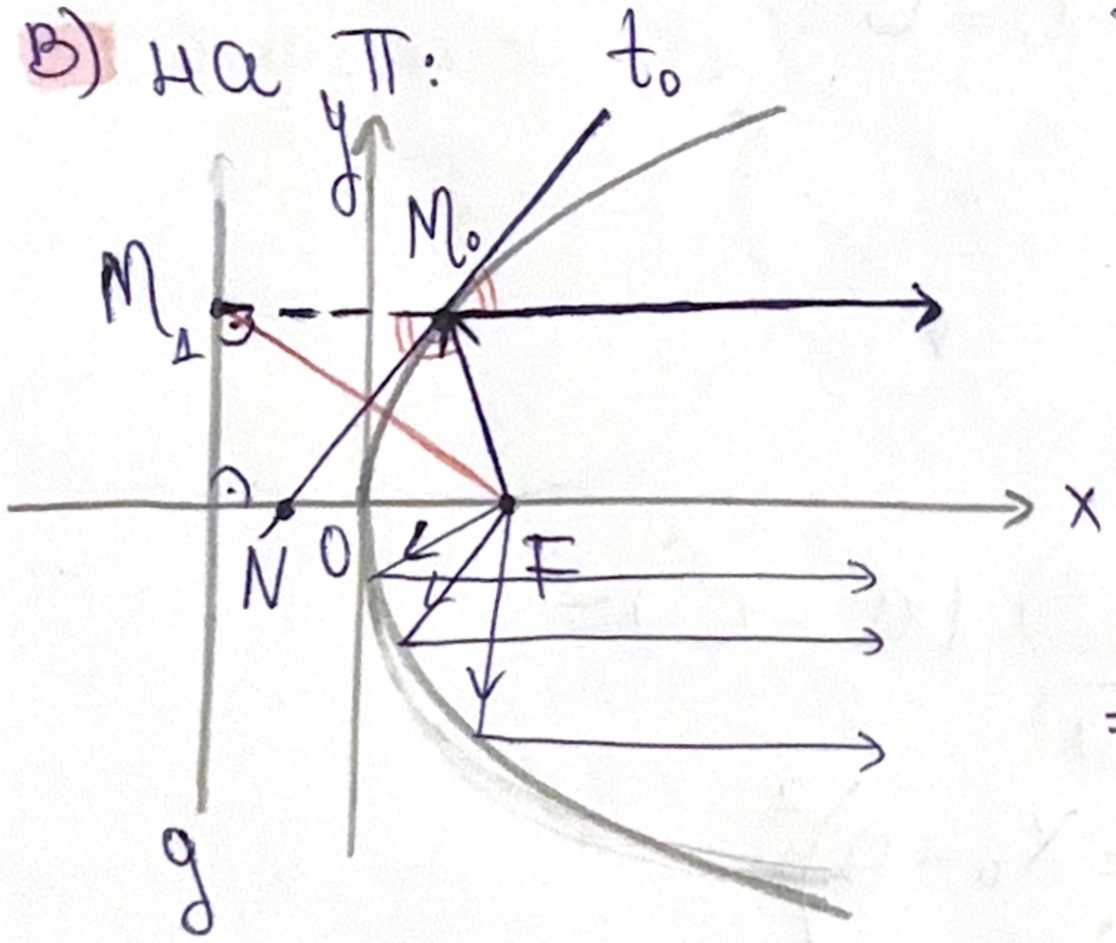
$N = t_0 \cap \chi$

$\Rightarrow N\left(\frac{a^2}{x_0}, 0\right)$

Правата t_0 е вътрешна стопоповеща на тела при върха M_0 на $\Delta F_1F_2M_0$, т.е. е изпълнено: $\frac{|\overrightarrow{F_1M_0}|}{|\overrightarrow{F_2M_0}|} = \frac{|\overrightarrow{F_1N}|}{|\overrightarrow{F_2N}|}$

(Аналог. на а))

в) на Π:



$\Pi: y^2 = 2px$ в ОКС, $K=Oxy$

$M_0(x_0,y_0) \in \Pi$

M_1 е ортогоналната проекция на M_0 в/у g

$g: x = -\frac{p}{2}$

$\Rightarrow M_1\left(-\frac{p}{2}, y_0\right)$

$t_0: y \cdot y_0 = p(x + x_0)$

Ще докажем, че t_0 е височина (стопоповеща и медиана) в ΔFM_0M_1, т.е. $t_0 \perp \overrightarrow{FM_1}$

док.: $F\left(\frac{p}{2}; 0\right)$

$M_1\left(-\frac{p}{2}, y_0\right)$ $\Rightarrow \overrightarrow{FM_1}(-p, y_0)$

$t_0: y.y_0 = p(x+x_0) \Rightarrow t_0: px - y_0.y + px_0 = 0$

$t_0 \parallel \vec{t_0}(-B, A) \Rightarrow \vec{t_0}(y_0, p)$

$(\vec{FM_1}.\vec{t_0}) = -p.y_0 + y_0.p = 0 \Rightarrow FM_1 \perp t_0 \Rightarrow$

$(\sphericalangle NM_0F = \sphericalangle M_1M_0N)$

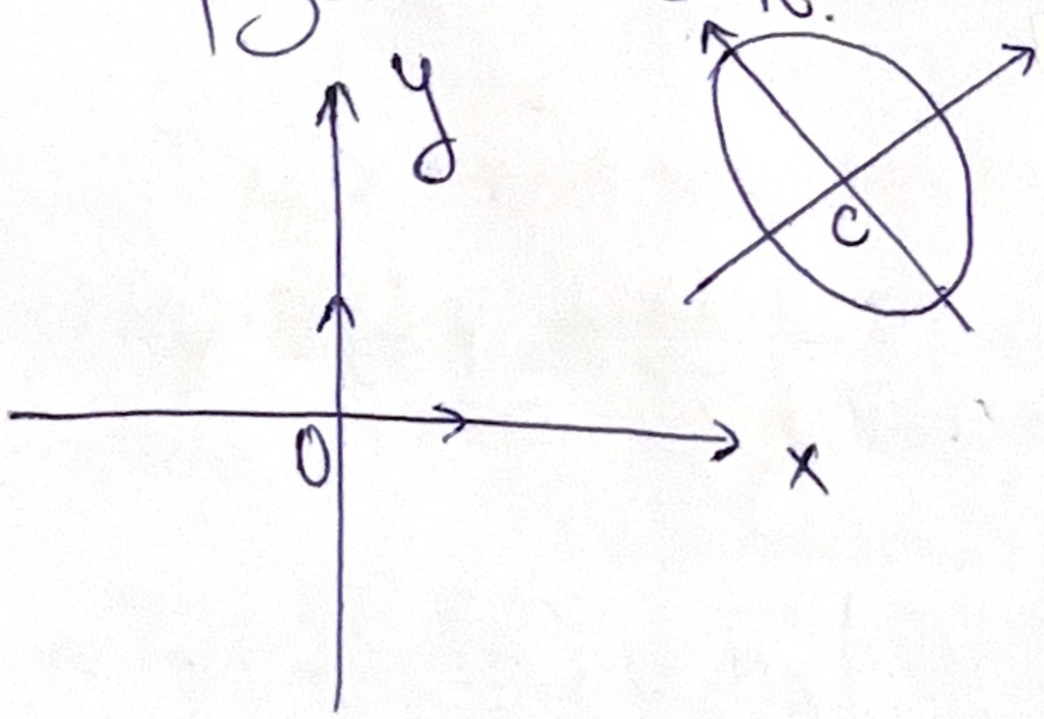
Метрични канонични у-я на кривите от 2-ра степен

1зад. ОКС $K = Oxy = O\vec{e_1}\vec{e_2}$

$k: 5x^2 + 8xy + 5y^2 - 18x - 18y + 9 = 0$

а) Да се намери метрично каноничено у-е на кривата k и последователните координатни трансформации, които водят до него.

→ Търсим нова ОКС, която е геометрично свързана с К.



$k: 5x^2 + 8xy + 5y^2 - 18x - 18y + 9 = 0$

$\overset{\shortparallel}{a_{11}} \quad \overset{\shortparallel}{2a_{12}} \quad \overset{\shortparallel}{a_{22}} \quad \overset{\shortparallel}{2a_{13}} \quad \overset{\shortparallel}{2a_{23}} \quad \overset{\shortparallel}{a_{33}}$

Разпределение матрицата:

$$A = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix}$$

Търсим нова векторна база, спрямо която А е в диагонален вид:

$A=\begin{pmatrix}5 & 4\\ 4 & 5\end{pmatrix} \longrightarrow A'=\begin{pmatrix}s_1 & 0\\ 0 & s_2\end{pmatrix}$

спрямо $\vec{e_1},\vec{e_2}$ спрямо $\vec{b_1},\vec{b_2}$

s_1 и s_2 са собствените стойности на A

$\vec{b_1}$ и $\vec{b_2}$ са собствените вектори на A

$\Rightarrow |A-s.E|=0 \rightarrow$ характеристично у-е на A

$$\begin{vmatrix}5-s & 4\\ 4 & 5-s\end{vmatrix}=0 \Rightarrow (5-s)^2-16=0$$

$5-s=\pm 4 \Rightarrow s_2=1$, $s_1=9$ → собствените стойности на A

1сл.: Ако $s_1.s_2>0$, то кривата k е от елиптичен тип, т.е. k има метрично каноничено у-е от вида $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

2сл.: Ако $s_1.s_2<0$, k е от хиперболичен тип, т.е. k има метрично каноничено у-е от вида $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

3сл.: Ако $s_1=0, s_2\neq 0$, k е от параболичен тип, т.е. k има метрично каноничено у-е от вида $y^2=2px$

Търсим собствените вектори на A:

$s_1 = 9 \Rightarrow \vec{b_1}(\alpha_1, \beta_1),\ |\vec{b_1}| = 1 \Leftrightarrow \alpha_1^2 + \beta_1^2 = 1$

$$(A - 9E)\begin{pmatrix}\alpha_1\\ \beta_1\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix} \Rightarrow \begin{pmatrix}-4 & 4\\ 4 & -4\end{pmatrix}\begin{pmatrix}\alpha_1\\ \beta_1\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix}$$

$\Rightarrow \begin{vmatrix} -4\alpha_1 + 4\beta_1 = 0 \\ \alpha_1^2 + \beta_1^2 = 1 \end{vmatrix} \Rightarrow \begin{vmatrix} \alpha_1 = \beta_1 \\ \alpha_1^2 + \beta_1^2 = 1 \end{vmatrix}$

$\Rightarrow \alpha_1 = \beta_1 = \pm \frac{1}{\sqrt{2}}$

$\vec{b_1} = \vec{b_1}$

(от $A.\vec{b_1} = s.\vec{b_1}$)

за $s_1 = 9 \Rightarrow \vec{b_1}\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

$s_2 = 1 \Rightarrow \vec{b_2}(\alpha_2, \beta_2),\ |\vec{b_2}| = 1 \Leftrightarrow \alpha_2^2 + \beta_2^2 = 1$

$$(A - 1E)\begin{pmatrix}\alpha_2\\ \beta_2\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix} \Rightarrow \begin{pmatrix}4 & 4\\ 4 & 4\end{pmatrix}\begin{pmatrix}\alpha_2\\ \beta_2\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix}$$

$\Rightarrow \begin{vmatrix} 4\alpha_2 + 4\beta_2 = 0 \\ \alpha_2^2 + \beta_2^2 = 1 \end{vmatrix} \Rightarrow \alpha_2 = -\beta_2 = \frac{-1}{\sqrt{2}}$

$\therefore$ за $s_2 = 1 \Rightarrow \vec{b_2}\left(-\frac{1}{\sqrt{2}}, +\frac{1}{\sqrt{2}}\right)$

$\Rightarrow \vec{b_1}\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$ и $\vec{b_2}\left(\frac{-\sqrt{2}}{2}, +\frac{\sqrt{2}}{2}\right)$

Извършваме смяна на ОКС:

$K = O\vec{e_1}\vec{e_2} \xrightarrow{T_1} K' = O_{\vec{b_1}\vec{b_2}}$: $Ox' \uparrow\uparrow \vec{b_1}$, $Oy' \uparrow\uparrow \vec{b_2}$

Нека т. $M(x,y)$ спрямо K и $M(x',y')$ спрямо K'

$$T_1:\begin{cases} x = \frac{\sqrt{2}}{2}x' + \left(-\frac{\sqrt{2}}{2}\right)y' \\ y = \frac{\sqrt{2}}{2}x' + \frac{\sqrt{2}}{2}y' \end{cases}$$

$\vec{b_1}$ $\vec{b_2}$

Спрямо $K' = O_{\vec{b_1}\vec{b_2}} = O_{x'y'}$ $A' = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow$

$a_{11} = 9, a_{12} = 0, a_{22} = 1 \Rightarrow$

У-ето на кривата k спрямо $K' = Ox'y'$ е:

$$k: 9(x')^2 + (y')^2 - 18\left(\frac{\sqrt{2}}{2}x' - \frac{\sqrt{2}}{2}y'\right) - 18\left(\frac{\sqrt{2}}{2}x' + \frac{\sqrt{2}}{2}y'\right) + 9 = 0$$

$$9(x')^2 + (y')^2 - 9(\sqrt{2}x' - \sqrt{2}y') - 9(\sqrt{2}x' + \sqrt{2}y') + 9 = 0$$

$$9(x')^2 + (y')^2 - 18\sqrt{2}x' + 9 = 0 \quad (\Delta)$$

У-ето, което получихме, не е метрично каноничено, защото т. O <u>не</u> съвпада с центъра C на кривата k.

Търсим $C(p, q)$.

Извършваме смяна на ОКС:

$K' = O_{x'y'} \xrightarrow{T_2} K'' = C_{x''y''}$: $C_{x''} \uparrow\uparrow O_{x'}$

$C_{y''} \uparrow\uparrow O_{y'}$

Нека $M(x', y')$ спрямо K' и $M(x'', y'')$ спрямо K''

$$T_2:\begin{cases} x' = x'' + p \\ y' = y'' + q \end{cases}$$

Заместване в (Δ):

$$9(x'' + p)^2 + (y'' + q)^2 - 18\sqrt{2}(x'' + p) + 9 = 0$$

$k: 9(x'')^2 + (y'')^2 + x''\underbrace{(18p - 18\sqrt{2})}_{=0} + \underbrace{2qy''}_{=0} + \underbrace{9p^2 + q^2 - 18\sqrt{2}p + 9}_{(*)} = 0$

$\begin{cases} 18p - 18\sqrt{2} = 0 \Rightarrow p = \sqrt{2} \\ 2q = 0 \Rightarrow q = 0 \end{cases} \Rightarrow C(\sqrt{2}, 0)$ спрямо K'

Пресмятаме $(*)$ - свободния член:

$(*) = 9.(\sqrt{2})^2 - 18\sqrt{2}.\sqrt{2} + 9 = 9.2 - 18.2 + 9 = -9$

$\Rightarrow k: 9(x'')^2 + (y'')^2 - 9 = 0$

$k: \frac{(x'')^2}{1^2} + \frac{(y'')^2}{3^2} = 1$

$a = 1, b = 3$

б) Да се намерят координатите на фокусите на елипсата k спрямо $K = Oxy$.

(1) коорд. спрямо $K'' = C_{x''y''}$ са

$F_1(0, -2\sqrt{2}), F_2(0, 2\sqrt{2})$ $\quad c^2 = b^2 - a^2 = 9 - 1 = 8$

спрямо K''

от $T_2: \begin{cases} x' = x'' + p = 0 + \sqrt{2} = \sqrt{2} \\ y' = y'' + q = -2\sqrt{2} + 0 = -2\sqrt{2} \end{cases} \Rightarrow F_1(\sqrt{2}, -2\sqrt{2})$ спрямо K'

$T_2: \begin{cases} x' = x'' + p = 0 + \sqrt{2} = \sqrt{2} \\ y' = y'' + q = 2\sqrt{2} + 0 = 2\sqrt{2} \end{cases} \Rightarrow F_2(\sqrt{2}, 2\sqrt{2})$ спрямо K'

ом T_1: $\begin{cases} x = \frac{\sqrt{2}}{2}x' - \frac{\sqrt{2}}{2}y' \\ y = \frac{\sqrt{2}}{2}x' + \frac{\sqrt{2}}{2}y' \end{cases}$

за $F_1 \rightarrow$ $x = \frac{\sqrt{2}}{2}\cdot\sqrt{2} - \frac{\sqrt{2}}{2}(-2\sqrt{2}) = 1 + 2 = 3$

$y = \frac{\sqrt{2}}{2}\cdot\sqrt{2} + \frac{\sqrt{2}}{2}(-2\sqrt{2}) = 1 - 2 = -1$

$\Rightarrow F_1(3,-1)$ спрямо K

за $F_2 \rightarrow$ $x = \frac{\sqrt{2}}{2}\cdot\sqrt{2} - \frac{\sqrt{2}}{2}\cdot 2\sqrt{2} = 1 - 2 = -1$

$y = \frac{\sqrt{2}}{2}\cdot\sqrt{2} + \frac{\sqrt{2}}{2}\cdot 2\sqrt{2} = 1 + 2 = 3$

$\Rightarrow F_2(-1,3)$ спрямо K

